

Group 8: Final Presentation

Single-Word Speech Intelligibility for Children with Repaired Cleft Lip and Palate

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Part I

The Consultation Process

How the project came together

- ▶ Emailed **Dr. Preisser** for a collaborator introduction
- ▶ Introduced to **Dr. David Zajac**
- ▶ Dr. Zajac needed 1–2 weeks to organize the dataset

Meeting #1: February 20, 2026

- ▶ **Before:** Shared collaborator agreement and timeline; reviewed project overview; set agenda
- ▶ **During:** Project overview; analytic decisions; data codebook
- ▶ **After:** Received dataset; wrote SAP

- ▶ **Before:** Drafted SAP; preliminary analysis; set agenda
- ▶ **During:** Reviewed SAP; explained LMM; previewed results
- ▶ **After:** Finalized analysis, tables, figures, and report

Challenges

- ▶ **Early scheduling delays:** Followed up proactively; sent materials in advance
- ▶ **Revised modeling plan:** Poor ICC in some listener groups undermined a mean SWI approach; switched to LMM on individual listener scores

Part II

The Study

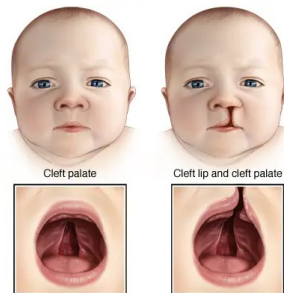
Background, design, models, and results

Study Background

Cleft Lip and Palate (CLP)

- ▶ Common birth defect (~1 in 1,031 births)
- ▶ Structural differences affect speech airflow
- ▶ Reduced intelligibility may persist after surgical repair

Speech Intelligibility: how well a listener understands a speaker (a key clinical outcome)



- ▶ **Aim 1:** Assess differences in overall speech intelligibility between children with CLP and children in the CP, OM, and TD groups
- ▶ **Aim 2:** Evaluate whether patterns of phonetic errors differ between children with CLP and children in the CP, OM, and TD groups

- ▶ Single-center, cross-sectional observational study; four groups: CLP, CP, TD, OM
- ▶ Children completed a 154-word speech test based on 11 phonetic contrasts
- ▶ Three adult listeners independently transcribed each child's recordings
- ▶ Outcome: % words correctly identified → **3 correlated scores per child**

Enrolled sample ($N = 115$ children)

- ▶ CLP ($n = 23$), CP ($n = 24$), TD ($n = 34$), OM ($n = 34$)
- ▶ Mean age 7.6 years; 60% male
- ▶ Mean intelligibility: 87.9% (range: 44.2%–98.1%)

Analytic sample ($n = 108$)

- ▶ 7 children with inadequate velopharyngeal (VP) status excluded (6 CLP, 1 CP)
- ▶ VP dysfunction is a physiological consequence of cleft palate; nearly fully confounded with CLP membership
- ▶ Full-sample analyses presented as sensitivity analyses

Preliminary Analysis

1. Inter-rater reliability

- ▶ ICC assessed within each of 22 listener groups
- ▶ Listener Groups 2, 3, and 7 had poor agreement ($ICC < 0.5$)
- ▶ $\Rightarrow$ LMM on individual listener scores preferred over mean SWI

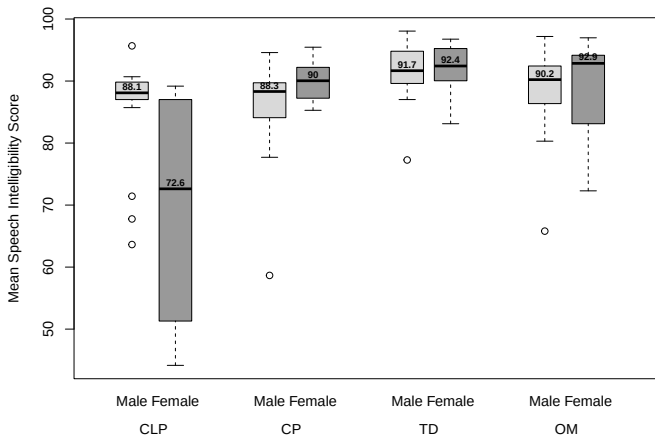
2. TD vs. OM comparability

- ▶ Bootstrapped TOST: observed difference = 3.08 points (90% CI: 0.87, 5.68)
- ▶ Formal equivalence not confirmed, but difference clinically negligible ($\sim 3\%$ of scale)
- ▶ $\Rightarrow$ TD and OM merged into a single **Control** group

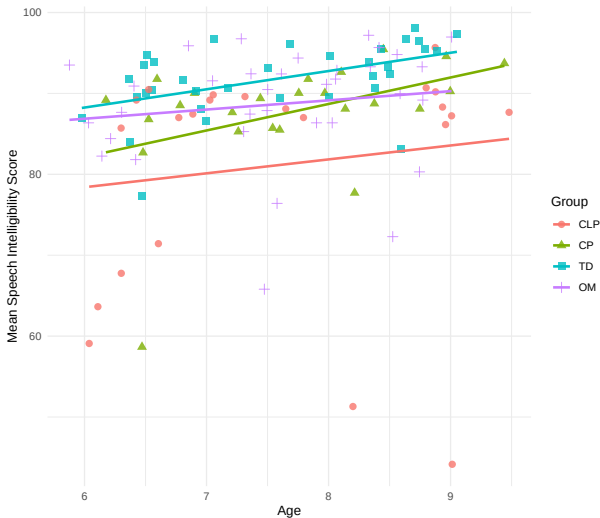
3. VP status

- ▶ CLP females with inadequate VP: mean SWI = 51.5 vs. 87.4 among adequate VP CLP females
- ▶ $\Rightarrow$ 7 children with inadequate VP excluded from primary and secondary analyses

Speech Intelligibility by Group and Sex



Speech Intelligibility vs. Age by Cohort



Aim 1: Overall Speech Intelligibility Differences

- ▶ Assess differences in overall speech intelligibility between children with CLP and children in the CP and Control (OM + TD) groups
- ▶ Analytic sample: $n = 108$ (adequate VP status only)

Aim 1: Analytical Approach

Method: Linear mixed model (LMM)

- ▶ Each child rated by 3 listeners → correlated observations
- ▶ Random intercept for child ID
- ▶ Reference group: CLP
- ▶ Covariates: age, sex
- ▶ Pairwise interactions screened via likelihood ratio test (LRT)
- ▶ No interaction retained (all LRT $p \geq 0.40$); contrasts pooled across sex
- ▶ Multiplicity: Hochberg adjustment at $\alpha = 0.05$

Aim 1: LMM

$$SWI_{ij} = \beta_0 + \beta_1 \text{Cohort}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + u_{ID(i)} + \varepsilon_{ij}$$

- ▶ No interaction terms retained (new_Cohort:Age $p = 0.677$; new_Cohort:Sex $p = 0.517$; Age:Sex $p = 0.398$)
- ▶ $u_{ID(i)}$: child-level random intercept (SD = 5.97); listener group variance = 0

Aim 1: Adjusted Cohort Contrasts

Contrast	Estimate	95% CI	Adjusted p
Control vs. CLP	+3.80	(-0.07, 7.67)	0.056
CP vs. CLP	+0.88	(-3.71, 5.47)	0.664

- ▶ Neither contrast reached statistical significance after Hochberg adjustment
- ▶ Control vs. CLP difference of 3.80 points approached significance ($p = 0.056$)
- ▶ Older age was positively associated with intelligibility ($p = 0.001$)

Aim 1: Fixed-Effect Estimates

Term	Estimate	95% CI	<i>p</i> -value
Intercept	70.18	(60.32, 80.03)	< 0.001
Control vs. CLP	+3.80	(−0.07, 7.67) [†]	0.056
CP vs. CLP	+0.88	(−3.71, 5.47) [†]	0.664
Age	+2.07	(0.84, 3.30)	0.001
Sex (Female)	+1.33	(−1.09, 3.76)	0.284

[†] Hochberg-adjusted 95% CI

- ▶ Older age significantly associated with higher intelligibility ($p = 0.001$)
- ▶ Sex was not a significant predictor; no interaction terms retained
- ▶ Random intercept SD: child ID = 5.97; residual SD = 2.68

Aim 2: Phonetic Error Patterns

- ▶ Evaluate whether patterns of phonetic errors differ between children with CLP and children in the CP and Control groups
- ▶ Same analytic sample ($n = 108$) and fixed effects as Aim 1

Aim 2: Analytical Approach

Method: 11 separate OLS regressions — one per phonetic contrast

- ▶ Same fixed effects as primary LMM (cohort, age, sex; no interactions retained)
- ▶ Reference group: CLP; contrasts pooled across sex

Multiple comparisons (Gatekeeping):

- ▶ **Family 1 (CLP vs. Control):** all 11 contrasts; Hochberg at $\alpha = 0.05$
- ▶ **Family 2 (CLP vs. CP):** only tested if Family 1 contrast is rejected

Aim 2: Regression Model

$$Y_{ik} = \beta_0 + \beta_1 \text{Cohort}_i + \beta_2 \text{Age}_i + \beta_3 \text{Sex}_i + \varepsilon_{ik}$$

- ▶ Y_{ik} : mean % correct for contrast k , child i
- ▶ Fit separately for each of 11 phonetic contrasts
- ▶ No interaction terms; VP status not included (analytic sample: adequate VP only)

Aim 2: All Phonetic Contrasts (Family 1)

Phonetic Contrast	Estimate	95% CI	Adj. p
Alveolar-palatal sibilant (see-she)	8.3	(2.1, 14.5)	0.066
Alveolar-velar stop (tap-cap)	2.7	(-2.2, 7.7)	0.941
Cluster-singleton (stop-top)	4.3	(-1.1, 9.6)	0.923
Initial consonant-null (pat-at)	1.7	(-2.0, 5.3)	0.941
Front-back vowel (fill-full)	3.0	(-2.6, 8.5)	0.941
Fricative-affricate (ship-chip)	5.0	(-1.3, 11.4)	0.923
High-low vowel (beat-bit)	4.4	(-0.2, 9.0)	0.591
Liquid-glide (ray-way)	7.0	(-2.1, 16.2)	0.923
Stop-fricative/affricate (base-vase)	3.8	(-2.0, 9.6)	0.941
Stop-nasal (bop-mop)	4.1	(-0.0, 8.3)	0.511
Voiced-voiceless stop (big-pig)	-0.6	(-5.7, 4.5)	0.941

Estimates = Control - CLP (mean % correct), adjusted for age and sex.
Family 2 not conducted for any contrast (all Family 1 adj. $p \geq 0.066$).

Aim 2: Results

No phonetic contrast was statistically significant after Hochberg adjustment

- ▶ All adjusted $p \geq 0.066$; Family 2 not conducted for any contrast

Largest estimated differences (Family 1, Control – CLP):

Phonetic Contrast	Estimate	95% CI
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Fricative-affricate	5.0	(-1.3, 11.4)

Estimates represent mean % correct difference (Control – CLP), adjusted for age and sex. None significant after Hochberg adjustment.

Conclusions

- ▶ Neither the Control vs. CLP nor the CP vs. CLP contrast reached statistical significance; the Control vs. CLP difference of 3.80 points approached significance (adjusted $p = 0.056$)
- ▶ Older age was positively associated with intelligibility across all groups ($p = 0.001$)
- ▶ At the phonetic level, no contrast-specific difference survived Hochberg adjustment; Family 2 was not conducted
- ▶ Findings suggest children with repaired CLP may have lower speech intelligibility than controls, though larger studies are needed to draw definitive conclusions

Thank You

We thank **Dr. David Zajac** for his collaboration, patience, and generosity in sharing both his data and his clinical expertise.

We also thank **Prof. John Preisser** for the introduction that made this project possible.

Questions?